

June 17, 2026

Dear Giovanni Neglia,

I am writing to express my strong interest in the PhD position on "Frugal Distributed Training with Volatile Resources" at Inria, located in Paris, France. As a passionate AI and Data Science Engineer, I am excited about the prospect of joining your esteemed institution and contributing to cutting-edge research in this field.

My background in data science and machine learning has equipped me with a strong mathematical foundation, programming skills, and experience with scalable training algorithms for distributed environments. My recent project on "Cardiovascular Medical Digital Twin" showcases my expertise in developing efficient inference models, such as *Medical_{Mamba}TS*, which leverages linear – time statespace modelsto enable sub – millisecond bedside inference.

I am particularly drawn to this PhD position because of its focus on frugal distributed training with volatile resources. My previous work on *SST_KODA_MMultiScaledemonstratesmyabilitytoenableefficientinferenceonresourceconstraineddevices* and *AdaptiveGates*. I believe that these skills will be highly relevant in developing novel training algorithms for distributed training on resource-constrained devices.

Furthermore, my experience with PyTorch Autograd in *PI_{DeepON}* has taught me the importance of physical constraints in machine learning models.

I am impressed by Inria's research excellence and its commitment to advancing digital science and technology. I am excited about the prospect of working with your team and contributing to innovative projects that have a significant impact on society.

Thank you for considering my application. I look forward to discussing this opportunity further.

Sincerely,

Ismail Aissaoui